Seahouses Harbour Soundscape

Recording Information

Location	Seahouses Harbour, Seahouses, Northumberland, UK
Coordinates	55.583379° N, –1.651591° W
Date & Time	14 January 2025, 15:34 GMT
Recording Equipment	Zoom F8n / Røde NT-SF1 ambisonic microphone
Recording Format	4-channel B-format (AmbiX, ACN/SN3D), 24-bit, 96 kHz

Processing & Render Details

The original 4-channel B-format ambisonic recording was decoded using a first-order UHJ stereo rendering method, optimised for headphone presentation. A 38-second excerpt was processed in REAPER 7.35 using the AmbiX VST plugin suite (Kronlachner, 2014). Spatial decoding used the ambix_binaural_o1 plugin with the icosahedron-3h3v preset. A gain adjustment of +4.1 dB was applied via JS: Volume Adjustment, followed by limiting via JS: Master Limiter (ceiling –1.2 dBTP). Final metering used Youlean Loudness Meter 2. Output rendered at 24-bit, 44.1 kHz, 2-channel stereo WAV (2067 kbps), 12 April 2025.

Processing Chain

DAW	REAPER 7.35
Ambisonic decoder	AmbiX VST — ambix_binaural_o1, icosahedron-3h3v preset
Gain adjustment	JS: Volume Adjustment (+4.1 dB)
Limiter	JS: Master Limiter (ceiling –1.2 dBTP)
Loudness meter	Youlean Loudness Meter 2
Render date	12 April 2025

Output Specification

Format	WAV, 24-bit PCM
Sample rate	44,100 Hz
Channels	2 (stereo)
Bit rate	2067 kbps
Duration	0:38.000

Loudness Metrics

Parameter	Value
Integrated Loudness (LUFS-I)	–16.9 LUFS
Short-Term Loudness max (LUFS-S)	–15.2 LUFS
Momentary Loudness max (LUFS-M)	–14.3 LUFS
Loudness Range (LRA)	3.9 LU
True Peak	–3.9 dBTP
Clipping	0 occurrences

Loudness normalisation target: –18 LUFS ±0.5 (ITU-R BS.1770-4 / AES TD1004.1.15-10).

Rationale

Selected for its representative ambient maritime character at Seahouses Harbour. The recording site offers unique comparative value within the dataset owing to its exposure to human and marine activity. Its inclusion allows for contrastive analysis against more tranquil settings, particularly those from the Leck Valley, contributing to the broader examination of contextual aural diversity.

REAPER Render Analysis

REAPER Render Sat Apr 12 10:33:31 2025

WAV: 24bit PCM, 44100Hz, 2ch, 2067kbps

File	Length	Peak	Clips	max LUFS-M	max LUFS-S	LUFS-I	LRA
Seahouses Harbour 38 sec.wav	0:38.000	-3.9	0	-14.3	-15.2	-16.9	3.9

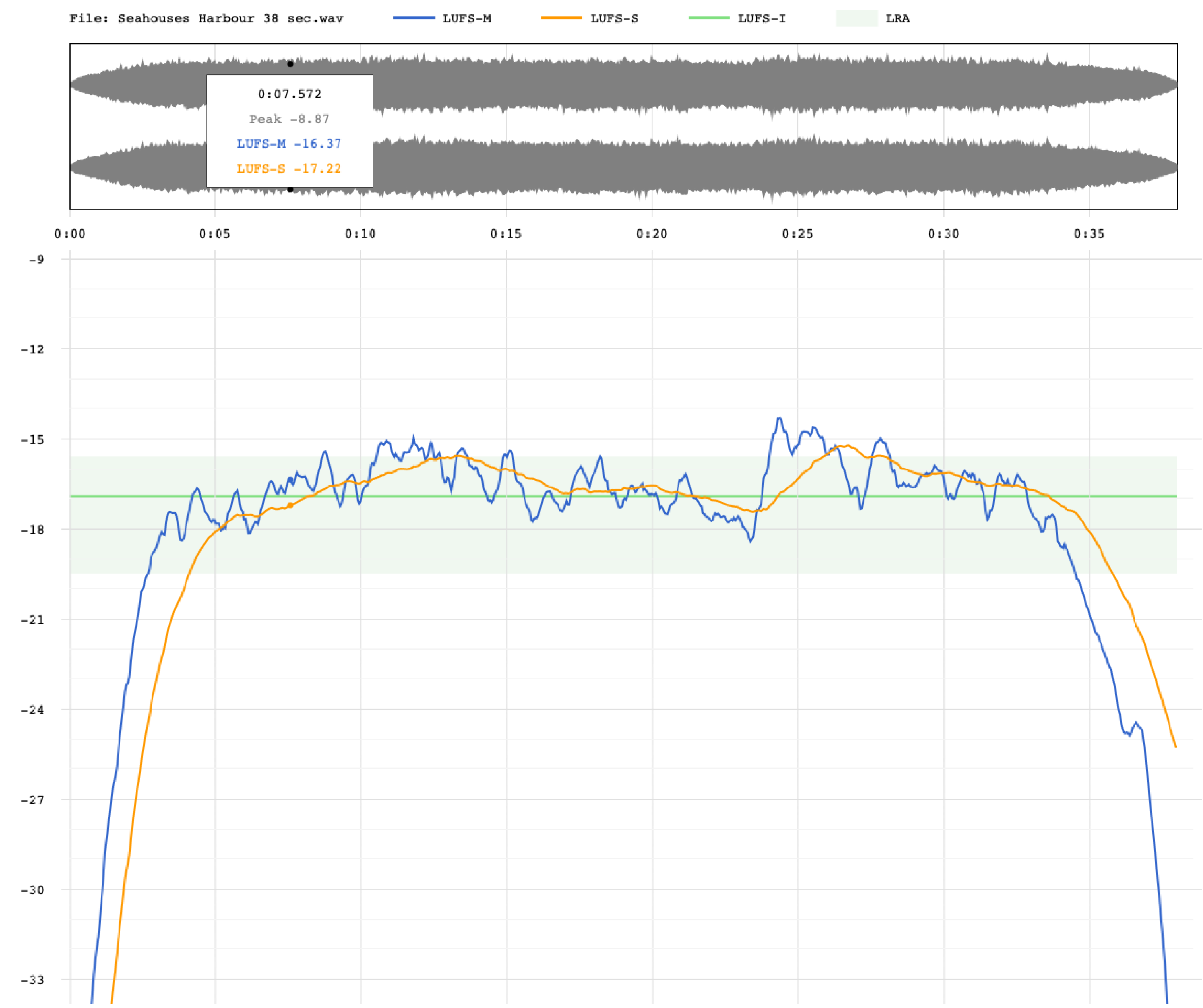

Figure: REAPER loudness render report — Seahouses Harbour Soundscape. Waveform display (top) and integrated loudness profile (LUFS-M blue, LUFS-S orange, LUFS-I green, LRA shaded region).